B

PUT

ROURKELA

PROJECT DOCUMENTATION REPORT

CUPGS Academic Feedback Manager

Anonymous Student Feedback System
with Role-Based Access Control

INSTITUTION

Centre for Urban & Planning
Graduate Studies (CUPGS)
BPUT Rourkela, Odisha

SYSTEM TYPE

Full-Stack Web Application
Progressive Web App (PWA)
Cross-Platform + Dark Theme

TECHNOLOGY STACK

React 18 | TypeScript | Node.js | Express
PostgreSQL | Drizzle ORM | Tailwind CSS | jsPDF

Academic Year 2024-25 | Version 1.0 | Confidential

ENGINEERING DEPARTMENTS COVERED

- Computer Science & Engineering (CSE)
- Electronics & Communication Engineering (ECE)
- Electrical Engineering (EE)
- Mechanical Engineering (ME)
- Civil Engineering (CE)

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1. Project Overview & Objectives

What is CUPGS Feedback Manager?

CUPGS Feedback Manager is a full-stack web application designed exclusively for the Centre for Urban and Planning Graduate Studies (CUPGS) at BPUT Rourkela. It enables students to submit anonymous feedback on courses and faculty, while providing Heads of Departments (HODs) with a real-time analytics dashboard and one-click PDF reports. The system enforces strict role-based access and anonymity guarantees.

WHAT IT IS

- Anonymous digital feedback system
- Drag-based half-star ratings (0.5 steps)
- HOD-customizable question forms
- Real-time analytics dashboard
- One-click PDF report generation
- Role-based access control (4 roles)

WHO USES IT

- Students — submit course feedback
- Faculty — view own performance data
- HOD — analytics + form customization
- Admin — institution-wide management

WHY IT WAS BUILT

- Replace paper-based feedback system
- Guarantee student anonymity
- Give HOD real-time actionable data
- Track faculty performance over time
- Improve transparency and trust

1.1. Core Objectives

Anonymous Feedback System
Students submit feedback without revealing their identity. A unique Reference ID (FB-YYYYMMDD-XXXXXX) is issued per submission. Faculty and HODs cannot identify which student gave which rating.

Half-Star Drag Rating
5 rating categories: Teaching Quality, Course Content, Lab Facilities, Study Material, and Overall. Each rated 0.5 to 5.0 in 0.5 increments using a drag interaction. Works on mouse and touch.

HOD Analytics Dashboard
HOD sees department-wide analytics: total feedback count, average ratings per course, category breakdowns, top and bottom performing courses, and a downloadable PDF report.

Customizable Feedback Forms
HOD can add custom questions (text, rating, or multiple-choice) to their department's feedback form from within the dashboard. No developer required.

Faculty-Course Integrity Check
The system verifies that a student can only rate a course that belongs to their department and that the feedback references a validly assigned faculty member, preventing cross-department manipulation.

Admin-Controlled Feedback Windows
Admin creates time-bound feedback windows (with start and end dates). Students can only submit feedback when an active window is open. Windows can be global or department-specific.

1.2. System Scope

IN SCOPE:

- 5 Engineering departments: CSE, ECE, EE, ME, CE
- Odd semester (July-October) and Even semester (November-June) feedback
- Academic year auto-detection based on current month (April = new year)
- Full audit trail via Reference IDs, no personally identifiable information stored

OUT OF SCOPE:

- Examination or results management
- Attendance and fee management
- Hostel or library systems
- External university integrations

1.3. Key Metrics

268+	5	4	5	100%	0.5
Total Courses	Departments	User Roles	Rating Cats.	Anonymous	Rating Step

1.4. Login Credentials & Academic Context

Login Credential Formats:

Student	Login ID: **CUPGS/DEPT/NNN (e.g. CUPGS/CSE/042)**	Password: **4-digit PIN (default: 1234)**
Faculty	Login ID: **CUPGS/DEPT/NNN (e.g. CUPGS/CSE/101)**	Password: **4-digit PIN**
HOD	Login ID: **HOD/DEPT/001 (e.g. HOD/CSE/001)**	Password: **DEPT@2025 (e.g. CSE@2025)**
Admin	Login ID: **bput@admin2025**	Password: **Admin password**

Semester System:

ODD Semester (July - October)
1st, 3rd, 5th, 7th semester courses. Feedback window opens in October.

EVEN Semester (November - June)
2nd, 4th, 6th, 8th semester courses. Feedback window opens in June.

Academic Year Calculation:

If current month is April or later (month >= 4): Academic Year = currentYear-(currentYear+1) e.g., 2025-26
If current month is before April (Jan-Mar): Academic Year = (currentYear-1)-currentYear e.g., 2024-25

2. Problem Statement & Solution Approach

2.1. Problem Statement

Existing Problems at CUPGS Before This System:

- Paper-based feedback forms were time-consuming to distribute, collect, and manually aggregate
- Students had no anonymity guarantee on paper — fear of identification led to dishonest or skipped responses
- HODs had to spend weeks manually tallying ratings and summarizing results — no real-time insight
- No course-by-course or category-by-category comparison was possible — no trend data across feedback period

2.2. How CUPGS Feedback Manager Solves Each Problem

PROBLEM 1:
Paper forms — no digitization

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SOLUTION:
100% digital web-based form submission, stored instantly in PostgreSQL, accessible from any device

PROBLEM 2:
No anonymity — fear of identification

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SOLUTION:
Reference ID system: only the student knows their code. Faculty and HODs cannot link a rating to a student

PROBLEM 3:
Manual aggregation takes weeks

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SOLUTION:
Real-time HOD dashboard with department analytics. One-click PDF report generation for download

PROBLEM 4:
No course-wise comparison possible

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SOLUTION:
Per-course rating breakdown across 5 categories with averages, sorting, and semester history

PROBLEM 5:
One standard form for all departments

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SOLUTION:
HOD form builder lets each HOD add custom text, rating, or multiple-choice questions for their dept

PROBLEM 6:
No control over submission timing

->

SOLUTION:
Admin creates time-bound feedback windows with start and end dates. Students cannot submit outside windows

2.3. Unique Features Not Typically Found in Academic Feedback Systems

Half-Star Drag Rating
Users drag to set ratings in 0.5 increments (0.5 to 5.0), not just whole numbers. Designed for mouse and touch input. Provides higher data precision than integer-only systems.

Live HOD Analytics
HOD dashboard refreshes in real time. Shows department average, per-course rating comparison by all 5 categories, top and bottom performers, and trend data by semester.

Course-Faculty Integrity
The server checks that the submitted course belongs to the student's department and that the referenced faculty member is actually assigned to that course, preventing fraudulent cross-department submissions.

Reference ID Anonymity
Each submission receives a unique ID formatted as FB-YYYYMMDD-XXXXXX. Students verify their own response; nobody else can use this ID to trace them. No student ID is ever stored alongside feedback.

HOD Form Builder
HOD adds or removes custom questions directly from their dashboard. Supported field types: short text, long text, single-choice, and rating scale. Changes apply immediately to the student feedback form.

One-Click PDF Report
HOD clicks 'Download Report' and receives a fully formatted PDF with college header, semester info, all courses, category ratings table, and averages. Generated entirely client-side using jsPDF.

2.4. Feature Comparison: Traditional vs. CUPGS Feedback Manager

Feature	Traditional Paper System	CUPGS Feedback Manager
Submission Method	X Physical paper form + pen	OK Digital web form, any device or browser
Student Anonymity	X None — handwriting is recognizable	OK Guaranteed — reference ID only, no name stored
Result Availability	X 2-4 weeks of manual counting	OK Instant — real-time dashboard always current
Custom Questions	X Requires reprinting all forms	OK HOD adds in 1 click from the dashboard
Rating Granularity	X 1 to 5 whole numbers only	OK 0.5 to 5.0 in drag-based half-star steps
Report Generation	X Manual written or typed summary	OK Auto-generated PDF report with one click
Access Timing Control	X No restriction on timing	OK Admin-controlled windows with start/end dates
Data Storage	X Physical files — risk of loss	OK PostgreSQL database — permanent and queryable
Faculty Comparison	X Difficult, requires manual work	OK Instant side-by-side course and category comparison

3. System Architecture & Features

3.1. Three-Tier Architecture

TIER 1 -- PRESENTATION LAYER (Frontend)

React 18 + TypeScript + Vite + Tailwind CSS + shadcn/ui components
Pages: Login | Submit Feedback | HOD Dashboard | Admin Panel | Faculty View
PWA: Installable, responsive, works on mobile and desktop browsers

[REST API -- HTTP/JSON via Axios]

TIER 2 -- APPLICATION LAYER (Backend API)

Node.js + Express.js + TypeScript | Runs on PORT 8080
Routes: /api/auth /api/feedback /api/departments /api/courses /api/faculty /api/windows
Middleware: Rate Limiting | Session Management | Input Validation (Zod) | Trust Proxy

[Drizzle ORM -- Type-safe SQL Queries]

TIER 3 -- DATA LAYER (Database)

PostgreSQL | 6 Tables | Replit-managed database via DATABASE_URL
Tables: departments courses faculty feedback form_templates feedback_windows
Notable: JSONB for custom answers, real type for 0.5-step rating precision

3.2. Frontend Pages & Routing

/ — **Login Page**
User selects their role (Student, Faculty, HOD, Admin), then enters ID and PIN. Successful auth redirects to the appropriate dashboard.

/submit-feedback — **Submit Feedback**
Student selects department and course, enters section and year, drags star ratings for 5 categories, adds optional comments, answers custom questions, and submits. Receives Reference ID on success.

/hod-dashboard — **HOD Dashboard**
HOD views department analytics, per-course rating table with 5-category breakdown, downloads PDF report, customizes feedback form, and checks feedback window status.

/admin — **Admin Panel**
Admin views institution-wide analytics for all 5 departments, manages courses and faculty records, creates and toggles feedback windows.

/faculty — **Faculty View**
Faculty sees performance data for their assigned courses: average ratings per category, anonymous student comments, and semester trends.

3.3. API Endpoints Overview

Method	Endpoint	Purpose
POST	/api/auth/login	Login for all roles (Student, Faculty, HOD, Admin)
POST	/api/feedback	Submit anonymous student feedback
GET	/api/feedback/check/:refId	Verify own submission via Reference ID
GET	/api/departments/:id/hod-report	Fetch HOD analytics data for department
GET	/api/departments/:id/form-template	Load custom feedback form template
POST	/api/departments/:id/form-template	Save HOD custom form configuration
GET	/api/windows	List all feedback windows (active status)
POST	/api/windows	Admin: create a new feedback window
GET	/api/analytics/dashboard	Admin: institution-wide analytics summary
GET	/api/courses	List courses with assigned faculty info

3.4. Rate Limiting & Security

[SEC] Session Management	Server-side express-session with SESSION_SECRET stored as an environment secret. Only a session cookie is stored in the browser — no passwords or tokens.
[FAST] Rate Limiting (3 Layers)	Global: 100 req/15 min per IP. Auth endpoint: 10 attempts/15 min. Feedback POST: 20 submissions/10 min. Trust proxy enabled for accurate IP detection.
[OK] Input Validation	Zod schema validation on every API route. Type checking, range enforcement, and required field checks are applied server-side. Clear error messages are returned on invalid input.
[LOCK] Anonymity Guarantee	Student login ID is never stored in the feedback table. Only the generated Reference ID is persisted. Even the database admin cannot link a feedback record to a student.
[SHD] Faculty-Course Integrity	Server verifies that the submitted course belongs to the student's department and that the referenced faculty is assigned to that course. Cross-department submissions are rejected.
[CAL] Window Enforcement	Feedback window status is checked server-side on every submission. Even if the client-side check is bypassed, the server rejects submissions outside active windows.

4. User Roles & Access Control

4.1. Role Overview

STUDENT	FACULTY	HOD (Head of Department)	ADMIN

STUDENT

ID: CUPGS/DEPT/NNN | PIN/Pass: 4-digit PIN (default: 1234)

CAN DO:
- Submit anonymous feedback for enrolled courses
- Use drag-based half-star rating (0.5 to 5.0 per category)
- Add optional text comments per rating category
- Answer HOD's custom questions if configured
- Receive a unique Reference ID after each submission
- Verify own submission using the Reference ID

CANNOT DO:
- View other students' feedback or ratings
- See faculty names linked to specific ratings
- Access HOD, Admin, or Faculty dashboards
- Submit feedback outside active feedback windows

FACULTY

ID: CUPGS/DEPT/NNN (same ID format) | PIN/Pass: 4-digit PIN

CAN DO:
- View analytics for own assigned courses only
- See average ratings per category across all feedback
- Read anonymous student comments (no identity revealed)
- Compare own performance across semesters
- View rating distribution and submission count

CANNOT DO:
- Submit any feedback
- Identify which student gave which rating
- Access another faculty member's data
- Access HOD or Admin dashboards

HOD (Head of Department)

ID: HOD/DEPT/001 | PIN/Pass: DEPT@2025 (e.g. CSE@2025 for CSE HOD)

CAN DO:
- View full department analytics dashboard
- See per-course rating breakdown across 5 categories
- Sort and compare all courses in the department
- Download a formatted PDF report with one click
- Add, edit, and remove custom feedback questions
- View current feedback window status and dates

CANNOT DO:
- Access individual student responses or identities
- View data from other departments
- Create or delete courses and faculty records
- Create or control feedback windows (Admin only)

ADMIN

ID: bput@admin2025 | PIN/Pass: Admin-configured password

CAN DO:
- View institution-wide analytics across all departments
- Manage departments (add, edit, update HOD info)
- Add, edit, and assign faculty to courses
- Create feedback windows (title, dates, dept or global)
- Toggle feedback windows open or closed at any time
- Compare cross-department performance metrics

CANNOT DO:
- View individual anonymous feedback content or comments
- Override a HOD's department-specific custom form configuration

4.2. Access Matrix Summary

5.1. Authentication Module

Authentication Module

Handles all user authentication with role detection and session management.

Step-by-Step Flow:

1. User selects their role from the login page (Student, Faculty, HOD, Admin)
2. User enters their ID and PIN or password
3. Server checks the ID format to determine role: CUPGS/* for students and faculty, HOD/* for HODs
4. PIN is verified against the stored record for that employee or student ID
5. On success, server creates a session; on failure, returns a clear error message
6. Rate limiting blocks IPs after 10 failed attempts within 15 minutes
7. Session persists until the browser is closed or the user logs out explicitly

5.2. Feedback Submission Module

Feedback Submission Module

Core module — collects, validates, and stores anonymous student feedback.

Step-by-Step Flow:

1. Student selects their department, then the specific course from a filtered dropdown
2. Faculty is auto-populated from the course record (or shown as 'Faculty TBA' if unassigned)
3. Semester and Academic Year are auto-calculated from the current date
4. Student drags star ratings for 5 categories, each from 0.5 to 5.0 in 0.5 steps
5. Optional text comments can be added for each rating category
6. HOD's custom questions are shown if configured; student fills them out
7. On submit: server validates inputs, checks window, verifies integrity, stores record, returns Reference ID

5.3. HOD Analytics & Dashboard Module

HOD Analytics & Dashboard Module

HOD's primary tool for real-time department performance analysis and reporting.

Step-by-Step Flow:

1. Dashboard loads with department summary: total feedback received and overall average rating
2. Per-course table shows 5-category rating breakdown for every course in the department
3. HOD can sort the table by any rating category to identify top or bottom performers
4. Trend section shows semester-on-semester comparison for selected courses
5. PDF Report button generates a formatted report with college header, all courses, and ratings
6. Form Builder section lets HOD add, reorder, or delete custom feedback questions
7. Window Status panel shows whether a feedback window is currently active, with its dates

5.4. Admin Institution Management Module

Admin Institution Management Module

Admin's complete control panel for institution-wide configuration and oversight.

Step-by-Step Flow:

1. Institution dashboard shows combined analytics summary for all 5 departments
2. Highest and lowest rated courses across the institution are highlighted
3. Course management: add new courses, assign or reassign faculty, set semester and academic year
4. Faculty management: add, edit, or deactivate faculty records with department assignment
5. Feedback Window creation: set title, description, start date, end date, and target department
6. Window toggle: activate or deactivate any window at any time with a single toggle switch
7. Cross-department comparison shows relative performance of all departments side by side

6. Technology Stack & Design System

6.1. Complete Technology Stack

FRONTEND STACK

React 18	Component-based UI library with hooks-based state management
TypeScript	Statically typed JavaScript — compile-time error detection across all files
Vite	Build tool with lightning-fast HMR and optimized production bundles
Tailwind CSS	Utility-first CSS framework for responsive dark-themed design
shadcn/ui	Accessible component library: buttons, cards, dialogs, form inputs
React Query	Server state management: caching, background refetch, loading/error states
React Hook Form	Form state management with validation, error messages, and submit handling
jsPDF + html2canvas	Client-side PDF generation: HOD report is built in the browser on download

BACKEND STACK

Node.js	JavaScript runtime with non-blocking I/O and event-driven architecture
Express.js	HTTP server framework for routing, middleware chaining, and session handling
TypeScript	Full type safety on the server, sharing types with the frontend codebase
express-session	Server-side session management to maintain authentication state between requests
express-rate-limit	API rate limiting: protects against brute-force attacks and request flooding
Zod	Schema declaration and validation library — parses and validates all API inputs
Pino	High-performance structured JSON logging for production-grade observability

DATABASE STACK

PostgreSQL	Production-grade relational database: ACID compliant, JSON support, scalable
Drizzle ORM	Type-safe ORM with a schema-first approach and SQL-like query syntax
Drizzle Kit	Migration tooling: schema changes are pushed safely to the database
JSONB columns	Used for feedback.custom_answers and form_templates.fields for flexible storage
real type	Rating columns use PostgreSQL real (float) type to support 0.5 precision ratings

INFRASTRUCTURE & TOOLING STACK

Replit	Cloud development and hosting platform; manages ports, secrets, and domains
pnpm workspaces	Monorepo setup — api-server and bput-feedback as separate workspace packages
PWA	Web App Manifest + Service Worker: installable on any device, no app store needed
Environment Secrets	SESSION_SECRET stored securely as a Replit environment variable, never in code

6.2. UI Design System

Dark Glassmorphism Theme

Background: #0f172a (Slate 900) | Cards use backdrop-blur with semi-transparent white overlay

Primary Blue: #1e3a8a | Accent: #2563eb | Success Green: #14532d | Danger Red: #dc2626

Background · Card BG · Primary · Accent · Success · Danger · Faculty · HOD · Light Text

7. How To Use The System (Step-by-Step Guides)

STUDENT -- Step by Step Guide

1. **Open the App**
   Navigate to the app URL in any browser, or launch the installed PWA
2. **Select Role**
   Choose 'Student' from the role selector on the login screen
3. **Log In**
   Enter ID (e.g. CUPGS/CSE/042) and 4-digit PIN, then click Login
4. **Select Course**
   Pick your department, then choose the specific course from the dropdown
5. **Rate Each Category**
   Drag the star control to set ratings in 0.5 steps for all 5 categories
6. **Add Comments**
   Optionally type text comments for each rating category
7. **Custom Questions**
   Fill in any custom questions the HOD has added for your department
8. **Submit**
   Click 'Submit Feedback' and copy the Reference ID shown on screen
9. **Verify (Optional)**
   Use the Reference ID on the verification page to confirm your submission

HOD (Head of Department) -- Step by Step Guide

1. **Log In**
   Enter ID (e.g. HOD/CSE/001) and password (e.g. CSE@2025)
2. **View Dashboard**
   Department analytics load automatically — total feedback count and overall average
3. **Analyze by Course**
   Review the per-course table showing 5-category rating breakdown for every course
4. **Sort and Filter**
   Click column headers to sort by any rating category — find top and bottom performers
5. **Download PDF**
   Click 'Download PDF Report' to get a fully formatted department report
6. **Customize Form**
   Use the 'Form Builder' section to add, reorder, or delete custom questions
7. **Check Window**
   Review the Feedback Window panel to see if a window is active and its date range

ADMIN -- Step by Step Guide

1. **Log In**
   Enter Admin ID (bput@admin2025) and the configured admin password
2. **Institution View**
   See the combined analytics summary for all 5 departments at a glance
3. **Manage Courses**
   Add new courses, assign faculty, set semester number and academic year
4. **Manage Faculty**
   Add or edit faculty records, assign to departments, deactivate as needed
5. **Create Window**
   Set a title, description, start date, end date, and target department (or global)
6. **Toggle Window**
   Flip the Active toggle on any window to open or close feedback collection
7. **Cross-Dept View**
   Use the comparison panel to see relative performance across all departments

FACULTY -- Step by Step Guide

1. **Log In**
   Enter Faculty ID (e.g. CUPGS/CSE/101) and 4-digit PIN
2. **View My Courses**
   The dashboard lists all courses assigned to you in the current semester
3. **Select a Course**
   Click on any course to see detailed feedback analytics for that course
4. **Review Ratings**
   See average ratings for each of the 5 categories and the overall average
5. **Read Comments**
   Anonymous student comments are displayed below the ratings — no names shown
6. **Track Progress**
   Switch between semesters to compare how your ratings have changed over time

8. Data Flow & Security Design

8.1. Feedback Submission Data Flow (10 Steps)

1. **Student submits** feedback form
2. **Frontend validates** input (required, ranges)
3. **POST /api/feedback** (HTTP JSON request)
4. **Rate limit check** (20 req per 10 min)
5. **Session auth check** (is student logged in?)
6. **Zod schema validation** (type + range check)
7. **Course-Faculty** integrity check
8. **Feedback window** check (is window open?)
9. **Store record in** PostgreSQL database
10. **Return Reference ID** (FB-YYYYMMDD-XXXXXX)

8.2. Database Schema -- All 6 Tables

departments

- id PK
- code UNIQUE
- name
- hod_name
- hod_employee_id
- created_at
- updated_at

faculty

- id PK
- department_id FK
- employee_id UNIQUE
- name
- designation
- email
- is_active

courses

- id PK
- department_id FK
- faculty_id FK nullable
- code UNIQUE
- name
- semester
- academic_year
- credits

feedback

- id PK
- reference_id UNIQUE
- course_id FK
- faculty_id FK?
- department_id FK
- semester
- academic_year
- 5 x real ratings
- comments text

form_templates

- id PK
- department_id FK UNIQUE
- fields JSONB
- created_at
- updated_at
- updated_by

feedback_windows

- id PK
- department_id FK nullable
- title
- is_active boolean
- start_date
- end_date
- description

8.3. Security Measures Implemented

Session Management	Server-side express-session. SESSION_SECRET is stored as an environment secret — never in code. The browser stores only a session cookie, no sensitive data.
Rate Limiting (3 Levels)	Global: 100 requests per 15 min per IP. Auth endpoint: 10 login attempts per 15 min. Feedback POST: 20 submissions per 10 min. Prevents brute-force and flooding.
Input Validation	Zod schema validation on every API route — type checks, value ranges, and required fields. Server returns field-level error messages; invalid input never reaches the database.
Anonymous Guarantee	Student login ID is never written to the feedback table. Only the system-generated Reference ID is stored. Even a database administrator cannot trace feedback to a student.
Integrity Enforcement	Server validates: course belongs to student's department, faculty is assigned to that course. Cross-department submissions are rejected at the server, independent of client-side checks.
Window Enforcement	Feedback window is verified server-side on every submission. Client-side bypass has no effect — the server checks the database for an active window before accepting any record.
No Raw SQL	All database access goes through Drizzle ORM with parameterized queries. SQL injection is effectively impossible — no string interpolation in queries.
Error Handling	Server errors are logged with Pino (JSON format). Clients receive generic error messages — stack traces and database details are never exposed in API responses.

8.4. Anonymity Architecture -- How It Is Guaranteed

How Anonymity Is Guaranteed:

When a student submits feedback, the server generates a cryptographically unique Reference ID (format: FB-YYYYMMDD-XXXXXX). This ID is stored in the feedback record. The student's login ID (e.g. CUPGS/CSE/042) is NEVER written to the feedback table. There is no column, no join, and no log entry that links a student's identity to their feedback.

STORED in the feedback table:
reference_id, course_id, dept_id, semester, year, ratings, comments, custom_answers

NEVER STORED in the feedback table:
student_id, employee_id, name, login credentials, IP address, browser fingerprint

- **Even the developer and database admin cannot trace feedback back to a specific student.**
- **Students can verify their own submission using their Reference ID — but no one else can use it to identify them.**

9. Summary & Future Scope

9.1. Project Summary

CUPGS Academic Feedback Manager is a complete full-stack web application built specifically for BPUT Rourkela's CUPGS engineering departments. Its primary goal is to allow students to submit honest, anonymous feedback on their courses and faculty — without fear of identification — and to give HODs immediate, actionable insights through a real-time analytics dashboard and professional PDF reports.

The system serves 4 distinct roles (Student, Faculty, HOD, Admin), enforces strict access boundaries, and stores all data securely in a PostgreSQL database. Anonymity is guaranteed by architecture: student identity is never stored alongside feedback records.

9.2. Key Deliverables Achieved

OK Anonymous Feedback System
Students submit via any device with a guaranteed Reference ID — no name stored

OK Half-Star Drag Rating
5 categories rated from 0.5 to 5.0 in 0.5 steps — drag interaction, mouse and touch

OK Real-Time HOD Dashboard
Department analytics with per-course rating breakdown, always up to date

OK One-Click PDF Report
HOD receives a fully formatted department report in seconds, no manual work

OK HOD Form Builder
Custom question types (text, rating, choice) added without any developer involvement

OK Faculty-Course Integrity
Server-side validation prevents cross-department or fraudulent course submissions

OK Admin Institution Control
Centralized management: departments, courses, faculty, windows — all in one panel

OK Feedback Window System
Time-controlled collection with date-bound windows that Admin opens and closes

OK Progressive Web App
Installable, responsive, splash screen, dark theme — works like a native app

OK Production-Grade Security
Rate limiting (3 layers), session management, Zod validation, no raw SQL exposure

9.3. Future Scope & Planned Enhancements

Phase 2 -- Short Term (next 3-6 months)

- Email notifications: HODs receive automated semester-end reports by email
- Charts in PDF report: bar charts and radar charts for visual rating comparison
- Student feedback history: view all past submissions using saved Reference IDs
- Bulk course import: import courses from an Excel/CSV file instead of one by one
- Multi-language support: Odia and Hindi language options for wider accessibility

Phase 3 -- Medium Term (6-18 months)

- AI sentiment analysis: automatic positive/negative/neutral tagging on student comments
- Semester trend graphs: visual performance tracking for faculty across multiple semesters
- Mobile app: React Native version for iOS and Android with push notifications
- Automated alerts: HOD receives auto-alert when a course rating drops below a threshold
- Comparative benchmarking: compare department ratings against similar institutions

Phase 4 -- Long Term (18+ months)

- Multi-institution network: extend the platform to cover multiple BPUT affiliated colleges
- NAAC/NIRF integration: feedback data exportable in formats accepted by accreditation bodies
- Predictive analytics: machine learning model to identify at-risk courses early in the semester
- Blockchain-verified anonymity: on-chain proof of anonymity for maximum student trust
- Faculty improvement plans: HOD can attach action items to low-rated courses, track resolution

9.4. Technical Best Practices Applied

- Monorepo architecture (pnpm workspaces) — frontend and backend co-located, shared types, single repository
- End-to-end type safety — TypeScript + Zod + Drizzle ORM eliminates an entire class of runtime type errors
- Separation of concerns — API route handlers, business logic, and database queries are kept in distinct layers
- Security by default — rate limiting, session management, and input validation are built in from day one, not added later

Project At A Glance